

Arrodes: Inferring Environmental Models through Multi-Robot Epistemic Belief Inference without Communication

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Abstract—In this paper we present a method to address the gap in collaborative exploration that arises when robots cannot share their latest measurements of and beliefs about the environment state. We propose *Arrodes*, an algorithm for inferring the internal world model of an agent exploring an environment through observations of its trajectory. Building off of Bayesian techniques for inferring beliefs in finite, discrete worlds, *Arrodes* uses a self-organizing map (SOM) as a finite, representative space of potential models to perform exact inference when observations are sparse. To handle the limitations of a finite inference space in expressing the potential variability of continuous environmental fields, such as novel transient upwellings in ocean current fields, *Arrodes* also extends to a generalized inference tier using empirical orthogonal functions (EOFs) to capture the continuous space of environment fields. We demonstrate both tiers - finite SOM-based inference and expressive EOF-defined inference - on a simulated AUV using model output around Ram’s Head on St. John in the U.S. Virgin Islands.

I. INTRODUCTION

Distributed exploration is a multi-robot paradigm for efficiently gathering informative samples of an environment as a group. As robotic agents determine the regions to gather samples from an internal environment model, effective distributed operation relies on the robots’ internal models being consistent with one another, and often necessitates communication of measurements and beliefs. Dynamic tasks like monitoring ocean currents pose a two-fold issue to teams, presenting a changing environment that strongly limits communication, so as robot beliefs evolve, they may be unable to share their insights [1], [2]. This renders teammates and human supervisors alike with an obfuscated understanding of robot behavior, unable to explain what is driving the robot’s choices. Without this understanding, team members risk wasting resources covering regions already observed, or worse, not detecting and reacting to novel transient phenomena. This leaves agents with a pressing question: when directly querying models and beliefs that drive a teammate’s behavior is infeasible, can observations of its actions reveal in reverse those same beliefs?

Attributing behavior to a rational underlying belief has also appeared in discussions of computational cognitive modeling and its extensions to human-robot interactions. Baker *et al.* [3] demonstrated that an approximately rational Bayesian model of an agent’s belief structure, termed Bayesian Theory of Mind (BToM), was sufficient to effectively capture the relation between belief and behavior. However, belief inference in this manner presents two difficulties. First, exploratory planners determine locations to sample using a transformation of environment belief that helps indicate regions with

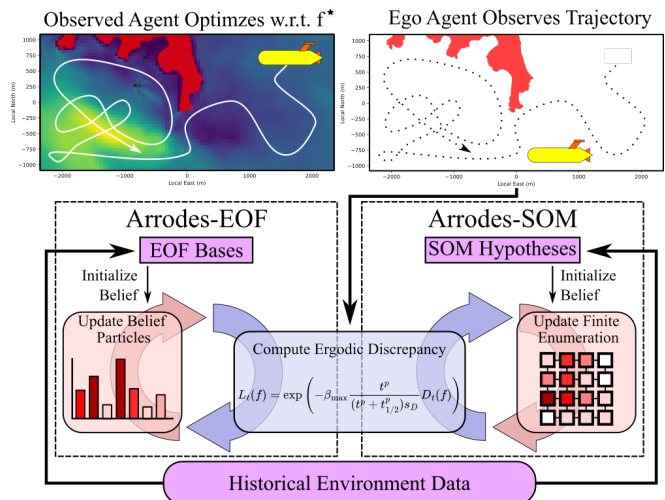


Fig. 1. Overview of Arrodes. (top left) The observed agent uses IPP to explore an environment using its environmental belief f^* . (top right) The ego agent observes the trajectory without access to f^* . (bottom) Arrodes constructs a hypothesis space from historical environmental data, either as a finite set of SOM fields or as a continuous hyperspace defined with an EOF basis. Arrodes evaluates the hypothesis space the best candidate field using the ergodic discrepancy between each candidate’s induced sampling field and the observed trajectory.

most promise of interesting information; observers must account for this transformation when reasoning how a sampling route arises from the underlying physical environment. The transformation also frequently prevents disambiguation, such as commonly seen in reinforcement learning between learned policies and their underlying reward functions [4]. Second, agents are often exploring with respect to continuous fields and gradients, which present a level of variability that simple discretizations of an environment’s state struggle to fully characterize, especially rare events. Contemporary Bayesian Theory of Mind (BToM) approaches that develop open-ended approaches to handle challenges of disambiguation still rely on having symbolically rich finite spaces of potential explanations [5], [6], a challenging structural requirement for environment models of continuous phenomena. The inference procedure needs a representation of potential environment beliefs that is tractable and sufficiently expressive, and should handle disambiguation from an exploration-centric transformation to the underlying environment model itself.

We answer this question in our paper with *Arrodes*, outlined in Fig. 1, which uses an inference system with two descriptive environment modeling frameworks to enable both classic and open-ended BToM methods for inferring the belief in the world an exploring agent has purely from observations of its sampling trajectory. The first tier of

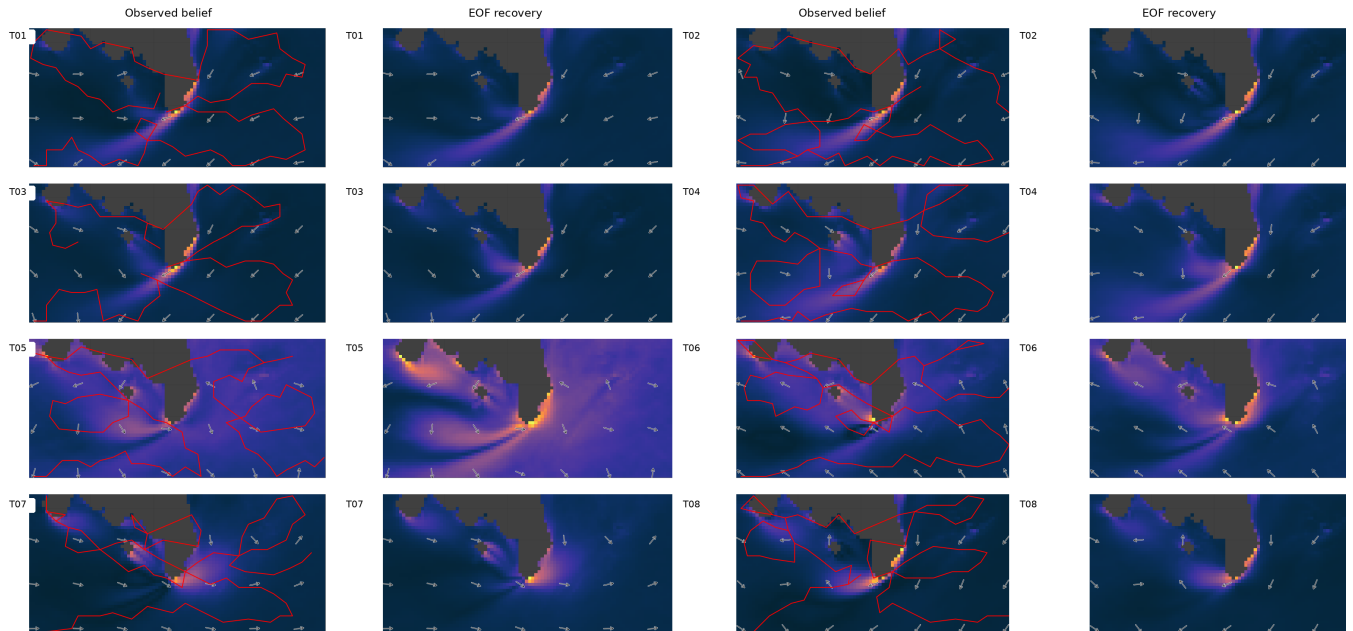


Fig. 2. Environmental field beliefs inferred from observed motion. Each horizontal pair of plots shows (left) the ground truth environment model of the observed agent, with its observable trajectory in red, followed by (right) Arrod's reconstruction using the EOF-based inferred posterior. The figures depict the spatial distribution of curl magnitude as a heatmap; arrows show the underlying flow for context.

Arrod's is finite mode inference, utilizing a self-organizing map (SOM) [7] to develop a set of states that sufficiently approximate the hypothesis space. The second tier presents an extension to open-ended inference, through decomposing the environment via empirical orthogonal function (EOFs) into an easily traversable hyperspace of potential world models [8]. To enable disambiguation between exploration reward functions and the underlying world, we model robots as ergodic with respect to the environment, since trajectories made by informative path planners are empirically as informative as ergodic trajectories [9], [10]. Enumerative BTOM methods are used for finite-mode inference, while open-ended inference uses probabilistic program proposals (SMCP³) [11] for inference in the EOF model hyperspace.

We evaluate Arrod's in the context of an exploration task by an Autonomous Underwater Vehicle (AUV) ROMS model output around Ram's Head, St. John, in the U.S. Virgin Islands [12]. We validate that our inference procedure is successful in recovering the observed agent's environment belief even as the divergence between the ego agent and the observed agent grows beyond multiple standard deviations. We show the effectiveness of the SOM approach is quickly converging on an approximate reconstruction of the observed agent's belief, requiring minimal observations of behavior, and further demonstrate that the expressiveness allotted by the EOF representation allows more accurate reconstructions after a sufficient number of behavior observations. We additionally show how open-ended inference with EOFs outperform finite-mode inference with SOMs when recovering out-of-distribution environment models formed from novel observations, and provide a qualitative metric for choosing between inference modes to use. To the best of our knowledge, our experiments mark the first time agents

attempt to infer beliefs in continuous environment fields through observations of exploratory behavior.

II. RELATED WORK

A. Distributed Exploration under Limited Communication

Informative path planning (IPP) optimizes exploration routes by planning a path to maximize an information gain metric. To prevent repeatedly sampling regions of diminishing gains, IPP approaches are typically adaptive, updating an environment model using observations and re-planning accordingly [1], [2]. In distributed multi-agent teams, local measurement updates lead to different models and can cause divergence in sampling plans. Existing approaches address this dependence through communication scheduling and rendezvous, explicitly modeling information availability through proximity, and through recasting planners as distributed optimization routines requiring consensus [13]–[15]. Recent approaches have taken greater attention to addressing issues of communication availability, but all concede that the dearth of a stable intra-agent network will result in plans degrading to effectively decoupled single-agent behavior.

In parallel to the stochastic IPP methods, ergodic planning supplies a useful alternative method for planning explorative paths by optimizing trajectories so robots spend time exploring regions proportional to the region's value [16]. Ergodic IPP considerably simplifies the communication requirements, but does not eliminate the need for communication altogether [17]. Subsequent work on ergodic planners improved robustness to noise and temporally varying fields [9], [18], but did not change the fundamental requirement of consistent environment models across agents for distributed exploration.

B. Belief Inference and Behavioral Explanation

Explaining the behavior of autonomous robots often draws from cognitive modeling of rational agents and from imitation learning. The latter, commonly leveraged in single- and multi-agent reinforcement learning applications, provides a predictor for teammate behavior that is broadly usable across a variety of robotic domains and system representations [19]. Recent work in inverse soft Q-learning and its multi-agent extensions avoids adversarial optimization while retaining the ability to recover policies and reward functions from demonstrations [4], [20]. However, imitation learning generally arrives at an intermediary representation between internal beliefs and behavior, such as a policy or Q-function, which serves as a powerful indicator of future actions and interests but does not explicitly expose the environmental beliefs driving those choices [21].

This distinction is particularly important in human-robot interaction, where anticipating the needs of teammates predicates effective cooperation, as observed by Kolb and Feigh's work on estimating human situation awareness using hand-crafted predicates and learned model reasoning [22]. Hellou further develops a structured approach to lifting inferred beliefs from behavior [23], using the Bayesian Theory of Mind (BToM) formulated by Baker *et al.* [3]. While these approaches rely on world models that do not adapt well to exploration of continuous environments, they elucidate the usability of probabilistic inference in recovering environment beliefs.

BToM methodology generally proposes that given an expressive and tractable method of proposing new hypotheses, observations can constrain and converge hypotheses about the internal states to a model would make an agent's behavior expected. Extensions such as Open-Ended Sequential Inverse Plan Search (SIPS) [24] support open-ended hypothesis spaces for BToM methods, while Han and Gmytrasiewicz extend BToM to interactive POMDPs for learning other agents' beliefs, preferences, and decision models [25]. Both assume a hypothesis search space that is either enumerable or comes with an efficient hypothesis proposal method, with support for probabilistic proposal networks developed in subsequent work [11]. In parallel, work on causal Bayesian inference of teammate beliefs in world construction and corresponding explanations for behavior [5], [6] demonstrate the applicability of BToM methods for deducing rationale for observations, but remained limited to symbolically-rich constructions of the world that easily supplied hypotheses about environment beliefs. Application to the field of exploration requires specifying a *tractable* hypothesis space, i.e. the space of environment models potentially held by an agent, and how those models drive behavior.

III. PROBLEM FORMULATION

We consider a multi-robot exploration mission with two agents, one is the observed agents, which moves through the spatial domain $\mathcal{X} \in \mathbb{R}^2$ be the spatial domain guided by an environmental field estimate $f^* \in \mathcal{F}$, where $\mathcal{F}$ is the set of all possible environmental fields. The other agent is the ego

agent, which is able to make observations of the *trajectory prefix* of the observed agent which is the locations it visits, $x_q \in \mathcal{X}$, and their dwell times, d_q ,

$$z_{1:T} = \{(x_q, d_q)\}_{q=1}^T. \quad (1)$$

The ego agent also knows the exploration objective $r : \mathcal{F} \times \mathcal{X} \rightarrow \mathbb{R}_{\geq 0}$ and, while it does not know the exact IPP method used to optimize the observed agent's behavior with respect to its *generating field*, f^* , the ego agent assumes that the observed agent allocates coverage ergodically with respect to the resulting sampling target.

The exploration objective assigns sampling value $r(f, x)$ to location x under a candidate field f . At locations $\xi_1, \dots, \xi_M$ with spatial weights w_m , it induces probability masses

$$\mu_m(f) = \frac{w_m r(f, \xi_m)}{\sum_{j=1}^M w_j r(f, \xi_j)}. \quad (2)$$

We call $\mu(f)$ the *sampling target*. Arrodes infers

$$p_t(f) = p(f \mid z_{1:t}, r), \quad \hat{f}_t = \mathbb{E}_{p_t}[f], \quad \hat{\mu}_t = \mathbb{E}_{p_t}[\mu(f)]. \quad (3)$$

The posterior mean $\hat{f}_t$ is the reconstructed field shown in Fig. 2; $\hat{\mu}_t$ describes the corresponding posterior expectation over sampling targets. **The goal of Arrodes, then, is to minimize the error between the posterior mean $\hat{f}_t$ and the generating field, f^* .**

IV. ENVIRONMENTAL MODELS

Arrodes uses two methods to reduce $\mathcal{F}$ to a tractable space for inference. Self-organizing maps (SOMs) define a discrete hypothesis set, while empirical orthogonal functions provide a fixed-size basis set that can describe a continuous hypothesis space.

A. Finite SOM Model

A SOM organizes learned environmental states on a graph $\mathcal{G} = (V, E_G)$. During training, the closest prototype and its graph neighbors move toward each sampled environmental state [7]. Each learned state defines a field $s_k \in \mathbb{R}^G$ in the representation used for inference, where G is the number of spatial field values. The resulting family retains recurring environmental patterns and the neighborhood relationships among them. Arrodes uses

$$\mathcal{F}_{\text{SOM}} = \{s_1, \dots, s_K\} \quad (4)$$

as its finite hypotheses. A posterior on the vertices thus identifies the learned states consistent with the trajectory, with the posterior mean $\sum_k p_t(k) s_k$ a mixture of those states.

B. Continuous EOF Model

The continuous model uses an EOF-basis decomposition derived from work by Young *et al.* [8]. Let $f^{(1)}, \dots, f^{(H)} \in \mathbb{R}^G$ be fields from the prior environmental record, $\bar{f}$ their mean, $W = \text{diag}(v_1, \dots, v_G)$ the full-field spatial weight matrix, and

$$\bar{f} = \frac{1}{H} \sum_{h=1}^H f^{(h)}, \quad X = \begin{bmatrix} f^{(1)} - \bar{f}, \dots, f^{(H)} - \bar{f} \end{bmatrix}. \quad (5)$$

From the weighted singular value decomposition

$$W^{1/2}X = U\Sigma V^\top, \quad (6)$$

we retain R modes and set $\Phi = W^{-1/2}U_{:,1:R}$. The weighted orthogonality $\Phi^\top W\Phi = I_R$ gives the projection and reconstruction

$$\mathbf{a}(f) = \Phi^\top W(f - \bar{f}), \quad f_{\mathbf{a}} = \bar{f} + \Phi \mathbf{a}. \quad (7)$$

The ego agent supplies the Gaussian prior $p_0(\mathbf{a}) = \mathcal{N}(\mathbf{a}; \mathbf{a}_0, P_0)$. In the simulated trials, $\mathbf{a}_0 = \mathbf{0}$ reconstructs the mean environment and P_0 retains the correlated coefficient variation estimated from the prior record. Projecting each SOM state through (7) places the finite hypotheses and continuous particles in the same rank- R coefficient geometry. This coefficient projection allows direct comparison of the SOM posterior's environmental support with that of the continuous model.

V. BEHAVIOR-CONDITIONED INFERENCE

For each candidate field, Arrodes computes a sampling target using the known objective and compares it with the observed trajectory's spatial distribution. This discrepancy determines a posterior over SOM states or EOF coefficients, from which the field is reconstructed by posterior averaging. Algorithm 1 provides a summary of the computation for both the finite and continuous inference modes.

Algorithm 1 Arrodes environmental belief inference

Require: trajectory $z_{1:T}$, objective r , SOM states $\{s_k, \rho_k\}$
Require: EOF model (f, Φ, p_0) and behavioral score parameters
1: Compute the sampling target $\mu(s_k)$ for every SOM state
2: Draw EOF coefficient particles $\mathbf{a}^{1:N} \sim p_0$
3: **for** $t = 1, \dots, T$ **do**
4: Form the occupation measure of $z_{1:t}$
5: Evaluate $D_t(s_k)$ and normalize the finite SOM posterior
6: Evaluate $D_t(f_{\mathbf{a}})$ and update the EOF particle posterior
7: Resample and rejuvenate the EOF particles when required
8: Reconstruct $\hat{f}_t^{\text{SOM}}$ and $\hat{f}_t^{\text{EOF}}$
9: **end for**
10: **return** both posterior histories and reconstructed fields

A. Trajectory Discrepancy and Behavioral Evidence

The occupation measure of the observed trajectory prefix is

$$\nu_t = \sum_{q=1}^t \omega_q \delta_{x_q}, \quad \omega_q = \frac{d_q}{\sum_{j=1}^t d_j}. \quad (8)$$

An ergodic planner drives this measure toward its sampling target. We evaluate a candidate field with the squared maximum mean discrepancy

$$D_t(f) = \text{MMD}_k^2(\nu_t, \mu(f)). \quad (9)$$

For the Gaussian kernel $k_\ell(x, y) = \exp[-\|x - y\|^2/(2\ell^2)]$, the discrepancy has the explicit form [26]

$$\begin{aligned} D_t(f) = & \sum_{q,s=1}^t \omega_q \omega_s k_\ell(x_q, x_s) \\ & - 2 \sum_{q=1}^t \sum_{m=1}^M \omega_q \mu_m(f) k_\ell(x_q, \xi_m) \\ & + \sum_{m,n=1}^M \mu_m(f) \mu_n(f) k_\ell(\xi_m, \xi_n). \end{aligned} \quad (10)$$

The first term describes the observed coverage, the last describes the candidate target, and the cross term measures their spatial agreement. Nearby visits contribute through their kernel similarity, avoiding a hard partition into spatial bins while still allowing the allocation relationship to be expressed in the discrepancy measure.

A short prefix would have covered only part of its target and should exert less influence than a mature trajectory. Thus, Arrodes scales the behavioral energy and generalized likelihood

$$\mathcal{E}_t(f) = \frac{\beta_t}{s_D} D_t(f), \quad L_t(f) = \exp[-\mathcal{E}_t(f)], \quad (11)$$

with

$$\beta_t = \beta_{\max} \frac{t^p}{t^p + t_{1/2}^p}. \quad (12)$$

The scale s_D is the median pairwise target-measure discrepancy, calibrated using prior environmental fields for EOF inference and learned states for SOM inference. At observation t , the likelihood is evaluated against the complete prefix $z_{1:t}$.

B. Finite SOM Update

Let ρ_k be the prior probability of SOM state s_k . Since the hypothesis set is finite, Arrodes evaluates its posterior directly:

$$p_t(k) = \frac{\rho_k L_t(s_k)}{\sum_{j=1}^K \rho_j L_t(s_j)}, \quad \hat{f}_t^{\text{SOM}} = \sum_{k=1}^K p_t(k) s_k. \quad (13)$$

This update is exact in this finite mode. To align the initial preference of both formulations, the SOM states are projected to EOF coefficients $\mathbf{a}_k$ and assigned

$$\rho_k \propto \exp\left[-\frac{1}{2}(\mathbf{a}_k - \mathbf{a}_0)^\top P_0^{-1}(\mathbf{a}_k - \mathbf{a}_0)\right]. \quad (14)$$

This transformation does not affect the way the posterior can converge on learned SOM states.

C. Continuous EOF Update

The EOF formulation conditions the Gaussian coefficient prior on the same behavioral evidence,

$$\pi_t(\mathbf{a}) \propto p_0(\mathbf{a}) \exp[-\mathcal{E}_t(f_{\mathbf{a}})]. \quad (15)$$

This posterior can freely explore coefficient combinations absent from the finite model, while retaining structure from the EOF basis. Arrodes represents (15) using the open-ended

generative inference mechanism Gen [27], and uses paired forward and backward proposals to move particles between successive prefix-conditioned targets, following SMCP³ [11].

The proposal follows a local Gauss–Newton approximation. Let m_t be the locally optimized coefficient vector, $J_t = \partial\mu(f_{\mathbf{a}})/\partial\mathbf{a}|_{m_t}$, and $\mathbf{K}$ the spatial kernel matrix at the quadrature sites. The transport covariance is

$$C_t = \gamma \left(P_0^{-1} + \frac{2\beta_t}{s_D} J_t^\top \mathbf{K} J_t \right)^{-1}, \quad C_t = B_t B_t^\top, \quad (16)$$

where γ scales the covariance. Consecutive approximations define the affine map

$$T_t(\mathbf{a}) = m_t + B_t B_{t-1}^{-1}(\mathbf{a} - m_{t-1}). \quad (17)$$

The backward proposal applies T_t^{-1} . Writing $\tilde{\pi}_t = p_0 L_t$ for the unnormalized target, a transported particle $\mathbf{a}_t^i = T_t(\mathbf{a}_{t-1}^i)$ receives weight

$$\tilde{w}_t^i = w_{t-1}^i \frac{\tilde{\pi}_t(\mathbf{a}_t^i)}{\tilde{\pi}_{t-1}(\mathbf{a}_{t-1}^i)} |\det(B_t B_{t-1}^{-1})|. \quad (18)$$

Normalizing these weights corrects both the change in behavioral evidence and the volume change under transport. Thus, the local Gaussian approximation guides particle motion while the weights continue to the unnormalized (15). In particular, a short trajectory prefix may move the local approximation toward a field explaining only the portion of the domain visited so far.

When $N_{\text{eff}} = (\sum_i (w_t^i)^2)^{-1}$ falls below a fraction of the particle count, residual resampling allocates copies according to the weights. Metropolis–Hastings moves then propose coefficient alternatives and accept them using the posterior and proposal-density ratio. These moves preserve π_t while allowing replicated particles to explore distinct explanations. The environmental reconstruction is

$$\hat{f}_t^{\text{EOF}} = \bar{f} + \Phi \sum_{i=1}^N w_t^i \mathbf{a}_t^i. \quad (19)$$

Continuous inference incurs greater computation than finite enumeration, but supports recovery between and beyond the learned SOM states.

VI. SIMULATED OCEAN EXPLORATION

A. Environmental Record and Mission

We use two years of hourly surface-current output from a Regional Ocean Modeling System hindcast of an approximately 2 km² coastal region west of Ram Head, St. John, U.S. Virgin Islands [7], [28]. Retaining every third snapshot gives 5,680 environmental states. To focus sampling on rotational flow structures, each state is represented by the magnitude of vertical curl divided by its spatial mean. The first 80% of the time-ordered record determines a rank-10 EOF model. The next 568 states calibrate the coefficient covariance and discrepancy scale; the final 568 provide candidate generating fields. A 7 × 7 SOM trained on the velocity history supplies 49 finite states, with each velocity prototype transformed into the same curl-shape representation.

For a field f , the simulated objective uses $r(f, x) = |f(x)| + \epsilon$, with ϵ equal to 0.01 times the training-field standard deviation. The field values remain in their unit-mean curl-shape representation for reconstruction and RMSE. Only the quadrature masses in (2) are normalized for trajectory planning and discrepancy evaluation.

The ego prior is centered at $\mathbf{a}_0 = \mathbf{0}$, the EOF mean environment, with the correlated covariance estimated from the calibration record. Both formulations use 196 quadrature sites and a Gaussian kernel bandwidth of 0.2 in normalized domain coordinates. For every generating field, the observed agent follows a 100-location kernel-ergodic trajectory toward its sampling target. Arrodes receives those locations and their dwell times, but none of the environmental data available to the observed agent. Continuous inference uses 512 particles, an effective-sample-size threshold of 0.65 N , two rejuvenation steps, and $\beta_{\text{max}} = 125$, $t_{1/2} = 18$, and $p = 2$. The proposal uses eight local optimization steps and $\gamma = 1$. The planner uses 180 optimization iterations, unit dwell weights, and an arbitrarily chosen speed of 0.0625 in domain coordinates per second; temporal plots use cumulative path length divided by this speed.

B. Distance from the Finite Representation

The role of continuous inference becomes clearest as the generating field leaves the region available to the SOM posterior mean. Let $P_0 = B_0 B_0^\top$ and define prior-whitened coefficients

$$\mathbf{y}(\mathbf{a}) = B_0^{-1}(\mathbf{a} - \mathbf{a}_0). \quad (20)$$

For projected SOM coefficients $\mathbf{a}_1, \dots, \mathbf{a}_K$, let $Y = [\mathbf{y}(\mathbf{a}_1), \dots, \mathbf{y}(\mathbf{a}_K)]$. The coefficient region available to every posterior-weighted SOM reconstruction is the convex hull

$$\mathcal{H}_{\text{SOM}} = \{Y\boldsymbol{\lambda} : \boldsymbol{\lambda} \geq 0, \mathbf{1}^\top \boldsymbol{\lambda} = 1\}. \quad (21)$$

We measure a generating field’s departure from this region by

$$\delta_{\text{SOM}}(\mathbf{a}) = \min_{\boldsymbol{\lambda} \in \Delta_K} \|\mathbf{y}(\mathbf{a}) - Y\boldsymbol{\lambda}\|_2. \quad (22)$$

One unit of δ_{SOM} is one standard deviation in the correlated EOF prior geometry.

We compute (22) offline for all 568 candidate fields, then select five temporally separated fields near each of ten distances across the available range. Two planner starting positions generate different observable trajectories from every field, yielding 100 simulated trials. The group-mean distances range from 0.61 to 4.78 prior standard deviations. We report spatially weighted field RMSE; aggregate markers show the mean across ten trajectories at each distance and error bars show one standard deviation.

VII. RESULTS

A. Reconstructing the Observed World Belief

The inferred fields in Fig. 2 recover prominent circulation structures in the observed agent’s belief from its trajectory. Their location and extent explain the agent’s concentration of sampling effort. Fine structure is less recovered, but the

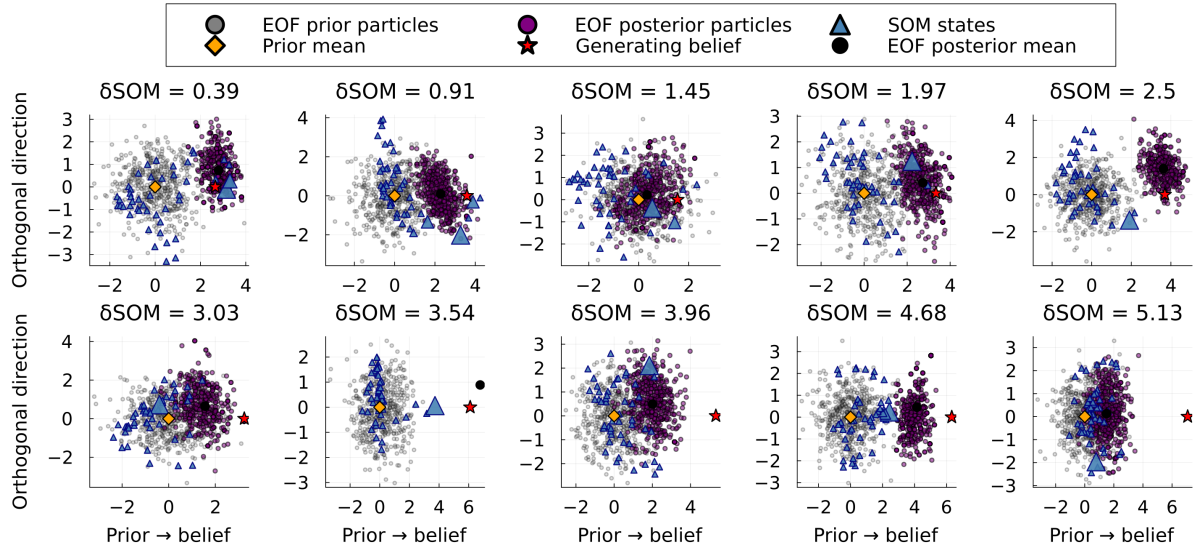


Fig. 3. The finite and continuous beliefs occupy the same EOF coefficient space. Across fields at increasing distance δ_{SOM} from the SOM region, the continuous particles move from the mean-environment prior toward fields whose sampling targets explain the observed trajectory. The SOM states delimit the region available to the finite posterior mean; each panel projects onto the direction from the prior mean to the generating belief and an orthogonal direction of posterior variation. Coordinates are in prior standard deviations; distances are evaluated in the full coefficient space. Triangle size indicates SOM posterior support.

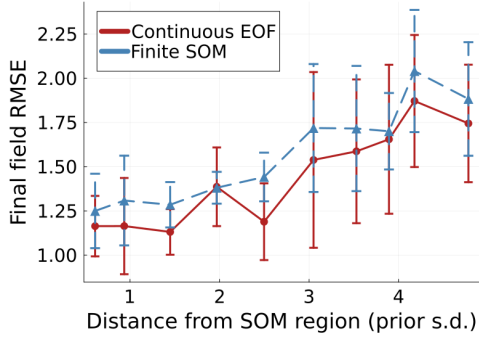


Fig. 4. Final field recovery as the generating field moves away from the region spanned by the SOM states. Markers show mean reconstruction error over repeated fields and starting positions at their actual mean distance; error bars show one standard deviation. EOF inference generally yields a closer reconstruction, with substantial variation across fields and starting positions.

reconstructions distinguish different environmental beliefs without access to the measurements that produced them.

Figure 3 shows how these reconstructions depart from the ego agent’s prior. The EOF posterior particles move away from the mean-environment prior toward the generating belief, while the SOM posterior selects among existing states. Their common coefficient coordinates show the extension afforded by EOF inference: reconstructed fields can lie beyond the region spanned by SOM mixtures.

B. Recovery beyond the SOM Region

The SOM reconstruction remains a useful approximation, but its error increases as generating fields depart from the learned states. The EOF formulation can account for some of this departure by adjusting coefficients beyond the SOM region.

EOF inference achieves lower mean final RMSE at nine of the ten distances in Fig. 4; the SOM mean is marginally lower near $\delta_{\text{SOM}} = 1.97$. Variation across generating fields and starting positions is substantial, so this is an overall improvement rather than uniform dominance. The expanded representation permits closer recovery of beliefs that the SOM can only approximate.

C. Trajectory Coverage Governs Continuous Recovery

This improvement requires a longer observation history. In Fig. 5, EOF field error initially rises while even the generating target has high discrepancy with the observed trajectory. Fields matching the incomplete route can therefore receive more support than the belief that actually generated it. As the route covers more of the target, discrepancy falls and both field and target reconstruction improve.

The histories in Fig. 6 show this behavior across the range of distances. SOM estimates often settle early, while EOF estimates undergo transient error increases before improving. The additional observations constrain continuous alternatives that a short route cannot distinguish.

The delay varies across fields: some EOF reconstructions improve rapidly after a sharp discrepancy decline, while others improve later in the route. The dominant association is thus between declining discrepancy and better EOF reconstruction, providing an empirical basis for choosing when its greater capacity becomes useful.

D. Choosing between SOM and EOF Inference

SOM inference is exact over its finite hypotheses, but approximate as a reconstruction of the observed agent’s field. Its computational advantage comes from enumerating a small set of complete states; its accuracy depends on whether those states describe the belief being inferred. EOF inference

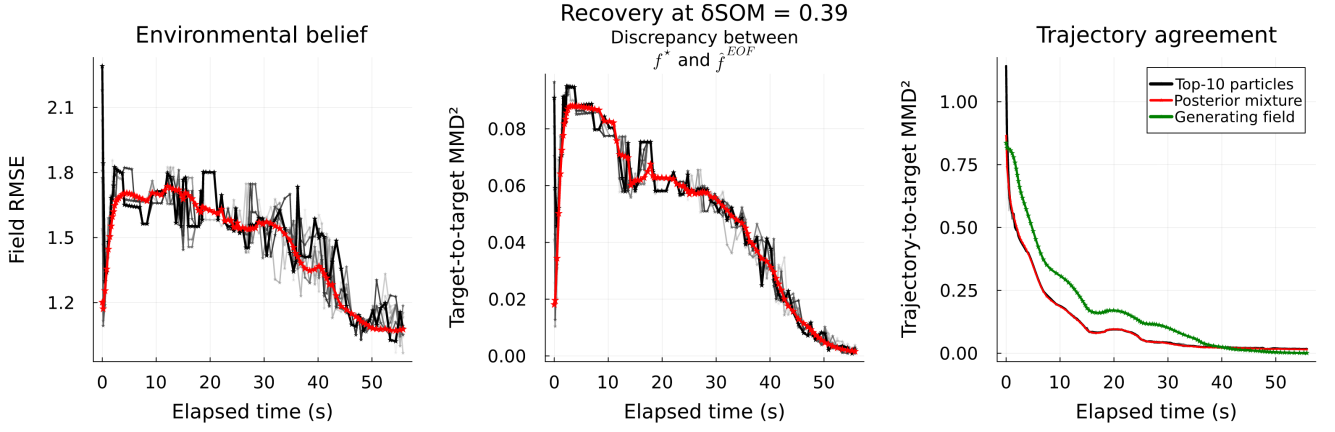


Fig. 5. Accuracy of EOF-basis field reconstruction over observation time. The left and center panels respectively show the RMSE of the environment field reconstruction and the target-to-target MMD² between the observed agent’s belief f^* and the highest-weight particles (gray-black) and the filter posterior (red). The right panel compares the discrepancy of the observed trajectory against the proposed fields of the highest-weight particles (gray/black), the filter posterior (red), and the observed agent’s belief (green). Reconstruction error is high for short observation times when the trajectory remains ergodically unrepresentative of underlying field, and then drops as increasing coverage gives greater evidence.

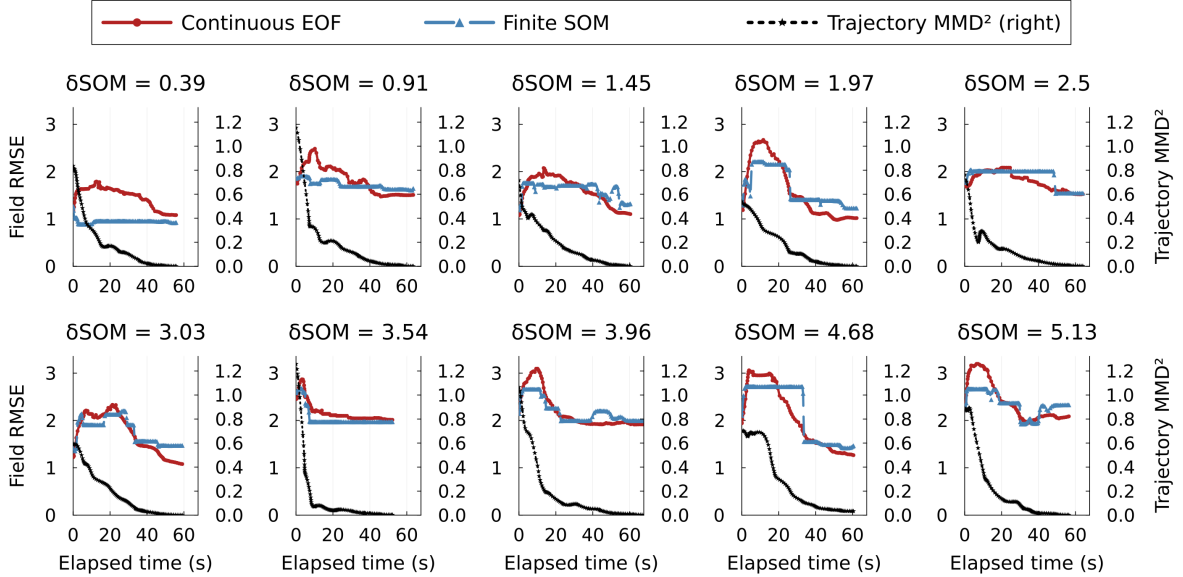


Fig. 6. Finite SOM and continuous EOF reconstruction histories across increasing distance from the SOM region. Colored curves show field RMSE on the left axis; black stars show trajectory-to-generating-target MMD² on the right. Markers indicate observations at elapsed physical times, and panel labels give δ_{SOM} in prior standard deviations. Finite reconstructions often settle early, whereas continuous recovery develops as the route covers the generating target. The generating target is used only for simulated evaluation.

relaxes this restriction by estimating continuous coefficients, at the cost of sampling a larger space of possible fields.

A short trajectory need not yet be ergodic with respect to the belief that generated it. EOF inference is particularly affected because it can construct alternative fields whose targets better match the visited locations. Longer observation histories reduce this ambiguity, allowing the continuous representation to improve on the finite approximation.

We therefore propose using the SOM reconstruction initially and treating a sharp decline in discrepancy as a signal to switch to EOF inference. The plotted discrepancy uses the known generating belief; an online rule must instead compare the trajectory with inferred targets. Because an incorrect field can also explain a short route, the reliability of that

observable indicator must be evaluated before the proposed switch can be used operationally.

Fields farther from the SOM region place greater demands on continuous inference, but distance alone does not determine convergence time. A field with separated sampling regions may require many observations before all are represented in the trajectory. The histories consequently support a dependence on both departure from the prior representation and the spatial evidence accumulated during exploration.

VIII. CONCLUSION

Arrodes extends belief inference to the environmental fields that guide exploratory sampling. SOM learned states provide a finite set of spatially resolved world hypotheses for exact Bayesian evaluation; EOF-basis decompositions

generalize these hypotheses to a continuous hyperspace that can represent beliefs outside the SOM's range. The simulated reconstructions show the benefits of utilizing both modes: SOM inference gives an efficient early approximation, while EOF inference recovers the observed agent's belief more accurately once enough of its trajectory has been observed. The relationship between recovery and declining ergodic discrepancy motivates switching between these approaches as evidence accumulates. This ability to reconstruct the model behind an exploration decision is foundational for robots to coordinate behavior and explain their behavior when their accumulated knowledge cannot be shared directly.

An AUV could use this inferred field to anticipate a teammate's sampling priorities and select complementary measurements; differences between their models could indicate regions that may contain novel information. For scientists aboard a support vessel, this reconstruction identifies the environmental assumptions behind a route; otherwise-anomalous repeated sampling around a novel feature becomes interpretable once that feature is visible in the inferred belief. Comparison with independent observations could then distinguish what is a scientifically useful response from a decision driven by an inaccurate onboard model.

The simulated experiments use static field beliefs, complete position traces, and a known curl-based sampling objective. They isolate inference of the environmental model from simultaneous uncertainty about goals or changes in the model during exploration. In the future we plan to extend Arródes to intermittent observations and evolving beliefs by accounting for when a trajectory was generated under each estimate. With the underlying belief of agents now explainable, the next step is to develop a distributed coordination system that can leverage this new understanding.

ACKNOWLEDGE OF USE OF AI TOOLS

Generative AI was used to generate portions of the code used in this paper. AI was not used to write any portion of the manuscript text, nor to generate any figures apart from writing code to generate plots from experimental data.

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